E-Commerce Sales & Customer Behavior Analysis

Exploratory Data Analysis (EDA) Project

- Dataset: UCI Online Retail (540,000+ Real Transactions)
- Tools: Python, Pandas, Matplotlib, Seaborn
- Focus: Data Cleaning, Visualizations, and Simple Business Takeaways

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Step 1: Checking the Raw Data & Issues Found

Real Data is Messy: The raw file had 541,909 rows, but many numbers were incomplete or strange.

Missing Values: About 25% of rows (135,080 rows) had no CustomerID. Also, 1,454 rows had missing descriptions.

5,268 Duplicate Rows: The same transaction was recorded twice, so I removed all 5,268 duplicates.

Cancelled Orders ('C'): Over 9,200 invoices started with 'C' and had negative quantities (customers returning items).

Negative Prices & Noise: Some prices were zero or negative (-£11,062), and there were test codes like 'POST' (postage) and 'M' (manual adjustments).

Cleaned Result: After cleaning, I got a clean dataset of 522,710 valid purchase records to analyze.

Step 2: What Are Customers Buying?

Cheap Gift Items: Most items in the store are inexpensive. The median price is just £2.08, and the median order line spend is £9.90.

Wholesale Quantities: The Quantity chart had clear spikes at 6, 12, 24, and 36 units. This shows that people buy in full boxes/packs to resell.

Top Revenue Product: 'REGENCY CAKESTAND 3 TIER' made the most money: over £174,000 in total sales!

Volume vs Revenue: The items with the highest quantity sold are cheap craft items (like toy gliders), but the real money comes from decorative home and gift items.

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Step 3: Where Do Orders Come From and When?

UK Dominates Volume: 91% of all transactions are from the United Kingdom.

The Netherlands Buys in Huge Bulk: While other European countries buy 10 to 12 items on average, the Netherlands buys a median of 72 items per order line! They are large wholesale buyers.

No Weekend Packing: Zero orders are recorded on Saturdays, showing the warehouse only operates Sunday to Friday.

Busiest Day & Peak Hours: Thursday has the highest number of orders. During the day, orders spike at 12:00 PM (lunch time) and 3:00 PM (afternoon).

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Step 4: Seasonality and the 80/20 Customer Rule

Christmas Holiday Spike: Sales stayed steady around £600k/month until August, but skyrocketed to £1.45 Million in November! Everyone buys gifts for the holidays.

The 80/20 Rule in Action: The top 10% of customers bring 61.4% of all revenue, and the top 20% of customers bring 74.6% of all revenue. A small group of loyal wholesale buyers keeps the business going.

High Cancellation Rate: Around 15% of all orders get cancelled or returned every single month. This wastes packing time and delivery costs.

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Step 5: Simple Recommendations & Conclusion

1. Reward Top 20% Customers: Give special discounts and dedicated support to the top 20% wholesale accounts because they bring 75% of total sales.

2. Order Stock in Summer for Q4: Since November sales double, order popular gift items from suppliers by July/August to prevent running out of stock.

3. Better Shipping for Dutch Buyers: Offer bulk freight discounts or a European hub for the Netherlands since they buy 72 units at a time.

4. Stop Out-of-Stock Orders: Show live stock numbers on the website so customers don't buy out-of-stock items and cancel later (cutting the 15% cancel rate).

5. What I Learned as a Beginner: EDA isn't just making pretty charts; it's about cleaning bad rows, asking simple questions, and turning data into clear decisions.